Aim:MANET Simulation with Directional Antenna
Simulate MANET using directional antennas and analyze improvements in spatial reuse and interference reduction.
Code:

```
# ======================================================
# MANET Simulation using AODV (NS-2.35 Compatible)
# ======================================================

set ns [new Simulator]

# Trace files
set tracefd [open manet.tr w]
$ns trace-all $tracefd

set namfd [open manet.nam w]
$ns namtrace-all-wireless $namfd 800 800

# Topography
set topo [new Topography]
$topo load_flatgrid 800 800

# Create GOD
create-god 5

# Wireless channel
set chan_1_ [new Channel/WirelessChannel]

# Node configuration
$ns node-config \
   -adhocRouting AODV \
   -llType LL \
   -macType Mac/802_11 \
   -ifqType Queue/DropTail/PriQueue \
   -ifqLen 50 \
   -antType Antenna/OmniAntenna \
   -propType Propagation/TwoRayGround \
   -phyType Phy/WirelessPhy \
   -channel $chan_1_ \
   -topoInstance $topo \
   -agentTrace ON \
   -routerTrace ON \
   -macTrace ON \
   -movementTrace ON

# Create nodes
for {set i 0} {$i < 5} {incr i} {
   set node_(i)[ns node]
}
```

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```
# Initial positions
$node_(0) set X_ 100
$node_(0) set Y_ 100

$node_(1) set X_ 250
$node_(1) set Y_ 200

$node_(2) set X_ 400
$node_(2) set Y_ 300

$node_(3) set X_ 550
$node_(3) set Y_ 400

$node_(4) set X_ 700
$node_(4) set Y_ 500

for {set i 0} {$i < 5} {incr i} {
    node_(i) set Z_ 0
    $ns initial_node_pos node_(i) 20
}

# UDP Agent
set udp0 [new Agent/UDP]
$ns attach-agent $node_(0) $udp0

# Null Agent
set null0 [new Agent/Null]
$ns attach-agent $node_(4) $null0

# Connect
$ns connect $udp0 $null0

# CBR Traffic
set cbr0 [new Application/Traffic/CBR]
$cbr0 set packetSize_ 512
$cbr0 set interval_ 0.1
$cbr0 attach-agent $udp0

# Mobility
nsat1.0"node_(1) setdest 300 300 10"
nsat2.0"node_(2) setdest 450 350 10"
nsat3.0"node_(3) setdest 600 450 10"

# Start traffic
nsat1.0"cbr0 start"
nsat8.0"cbr0 stop"

# Finish procedure
proc finish {} {
```

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```
    global ns tracefd namfd

    $ns flush-trace

    close $tracefd
    close $namfd

    exec nam manet.nam &
    exit 0
}

$ns at 10.0 "finish"

puts "Simulation Started..."
$ns run
```

OUTPUT:

